

## APPENDIX A. TRAINING AND IMPLEMENTATION DETAILS

## Training loop summary

## Algorithm A1 Physics-aligned MoE training

Input: history windows X, graph sequence A, targets Y, primary regime anchors R, auxiliary wake labels W (WTB only), model parameters  $\theta$

for each minibatch do

Build node-level graph context from the precomputed graph tensors

Encode X with the directed-diffusion GRU to obtain node context  $\mathbf{h}_t$

Compute gate logits Z and soft routing probabilities G

Compute expert forecasts and aggregate them with G

Evaluate L\_pred

If MoE mode is active:

add L\_bal on top-K routing statistics

add L\_align on valid primary regime anchors

if WTB: add L\_force on MPPT/pitch samples

if WTB: add L\_aux on wake labels

add L\_smooth on the final graph snapshot

Update theta with AdamW and mixed precision

end for

Select the checkpoint with the best validation RMSE

The wake weight is  $\tilde{\mathcal{A}}_t(i, j) = \exp(-d_{ij,t}^\parallel / \alpha) \exp(-|d_{ij,t}^\perp| / \beta) \mathbb{I}_{ij,t}^{\text{cone}}$ . If wind direction is missing, fall back to  $\mathcal{A}^{\text{static}}(i, j) = \exp(-\|\Delta \mathbf{p}_{ij}\|_2 / d_{\text{max}})$ . Only the strongest  $M$  inbound weights are retained. The wake score and flag are  $s_{ij,t}^{\text{wake}} = \sum_j \mathcal{A}_t(i, j)$  and  $\mathbf{w}_{i,t} = \sum_j \mathcal{A}_t(i, j) \mathbf{w}_j$ .

$$\mathbf{w}_{i,t} = \begin{cases} 1, & s_{ij,t}^{\text{wake}} \geq q_{0.75}^{\text{wake}} \wedge R_{i,t}^{\text{wtb}} \in \{1, 2\}, \\ 0, & s_{ij,t}^{\text{wake}} < q_{0.75}^{\text{wake}} \wedge R_{i,t}^{\text{wtb}} \in \{1, 2\}, \\ \emptyset, & \text{otherwise.} \end{cases}$$

For ERA5, the retained symmetric graph uses a Gaussian kernel on great-circle distances:

$$\mathcal{A}(i, j) = \exp\left(-\frac{d_{ij}^2}{2\sigma^2}\right) \mathbf{1}[j \in \mathcal{N}_{k_{\text{nn}}}(i) \text{ or } i \in \mathcal{N}_{k_{\text{nn}}}(j)],$$

where  $\sigma$  is the median retained neighbor distance on the training graph.

## Shared constants

## Model-family and supplementary evidence index

TABLE I  
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| Table A1. In-family model comparison used for mechanism validation. |                      |                                                                                                          |                            |                           |
|---------------------------------------------------------------------|----------------------|----------------------------------------------------------------------------------------------------------|----------------------------|---------------------------|
| Model                                                               | Routing              | Extra terms                                                                                              | Params                     | Role                      |
| Dense (matched)                                                     | Single dense head    | None                                                                                                     | WTB 110,012 / ERA5 103,272 | Shared-mapping reference  |
| Unconstrained MoE                                                   | Node-level soft gate | Prediction loss only                                                                                     | WTB 110,012 / ERA5 103,843 | Routed-capacity control   |
| Corrected routing comparator                                        | Node-level soft gate | WTB boundary-forced: $L_{\text{bal}} + L_{\text{align}} + L_{\text{force}}$ ; ERA5: full corrected stack | WTB 110,012 / ERA5 103,843 | Accountability comparator |

## Auxiliary losses

Let  $\mathcal{B}$  denote the routed samples after masks, with  $B = |\mathcal{B}|$ . The soft importance and normalized share of expert  $e$  are  $I_e = \frac{1}{B} \sum_{n=1}^B g_n^{(e)}$  and  $P_e = I_e / \sum_r I_r$ . The top-K load is  $f_e = \frac{1}{B} \sum_{n=1}^B \mathbf{1}[e \in \text{TopK}(\mathbf{g}_n)]$ , giving

$$\mathcal{L}_{\text{bal}} = E \sum_{e=1}^E f_e P_e - 1.$$

For WTB wake supervision,

$$\mathcal{L}_{\text{aux}} = \frac{1}{|\Omega_{\text{wake}}|} \sum_{(i,t) \in \Omega_{\text{wake}}} \text{BCE}(z_{i,t}^{(\text{wake})}, W_{i,t}),$$

where  $\Omega_{\text{wake}}$  contains only MPPT and pitch-control samples with defined wake flags. The graph-smoothness penalty is

$$\mathcal{L}_{\text{smooth}} = \frac{\sum_{t \in \mathcal{T}_{\mathcal{B}}} \sum_{i,j} \mathcal{A}_t(i, j) \|\mathbf{g}_{i,t} - \mathbf{g}_{j,t}\|_2^2}{\sum_{t \in \mathcal{T}_{\mathcal{B}}} \sum_{i,j} \mathcal{A}_t(i, j) + \epsilon}.$$

## Graph construction details

For WTB, the downstream unit vector is  $\mathbf{u}_{i,t} = [\sin(\theta_{i,t} + \pi), \cos(\theta_{i,t} + \pi)]^\top$ . With  $\Delta \mathbf{p}_{ij} = \mathbf{p}_j - \mathbf{p}_i$ , the streamwise and cross-stream distances are  $d_{ij,t}^\parallel = \Delta \mathbf{p}_{ij}^\top \mathbf{u}_{i,t}$  and  $d_{ij,t}^\perp = |\Delta p_{ij}^x u_{i,t}^y - \Delta p_{ij}^y u_{i,t}^x|$ . A candidate edge activates when

$$\mathbb{I}_{ij,t}^{\text{cone}} = \mathbf{1}\left[d_{ij,t}^\parallel > 0 \wedge \arctan\left(\frac{d_{ij,t}^\perp}{\max(d_{ij,t}^\parallel, 10^{-6})}\right) \leq \phi\right].$$

TABLE II  
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Table A3. Shared architecture, graph, and training constants used in the reported experiments.

| Item                                      | WTB                | ERA5                            | Role                                      |
|-------------------------------------------|--------------------|---------------------------------|-------------------------------------------|
| History / horizon ( $H, P$ )              | (36, 24)           | (36, 24)                        | Input and forecast lengths                |
| Gate softmax temperature $\tau$           | 0.7                | 0.7                             | Soft routing sharpness                    |
| Top-K in $L_{\text{bal}}$                 | 1                  | 1                               | Balancing statistic only                  |
| Hidden size $d$                           | 64                 | 64                              | Shared encoder width                      |
| $k_{\text{nn}}$ candidates                | 8                  | 8                               | Initial neighborhood size                 |
| Maximum candidate radius $d_{\text{max}}$ | 1500 m             | not used                        | WTB proximity fallback / candidate filter |
| Wake-cone half-angle $\phi$               | 25°                | not used                        | WTB downstream activation                 |
| Streamwise decay $\alpha$                 | 1200 m             | not used                        | WTB wake attenuation                      |
| Cross-stream decay $\beta$                | 400 m              | not used                        | WTB wake attenuation                      |
| Retained inbound edges $M$                | 5                  | symmetric $k_{\text{nn}}$ graph | Final graph sparsity                      |
| Optimizer                                 | AdamW              | AdamW                           | Shared optimizer                          |
| Learning rate                             | $2 \times 10^{-3}$ | $2 \times 10^{-3}$              | Shared schedule                           |
| Weight decay                              | $10^{-4}$          | $10^{-4}$                       | Shared regularization                     |
| Batch size                                | 16                 | 16                              | Shared mini-batch size                    |
| Gradient clipping                         | 1.0                | 1.0                             | Shared training stabilization             |

### Dataset-specific thresholds and routing weights

TABLE III  
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**Table A4.** Dataset-specific regime thresholds used to construct routing anchors.

| Dataset | Threshold                   | Value and meaning                                                            |
|---------|-----------------------------|------------------------------------------------------------------------------|
| WTB     | $u_{\text{idle}}$           | 3.0 m s <sup>-1</sup> , idle / active split                                  |
| WTB     | $u_{\text{rated}}$          | 10.5 m s <sup>-1</sup> , MPPT / pitch boundary                               |
| WTB     | $p_{\text{th}}$             | 2.0 deg, pitch-angle threshold                                               |
| WTB     | $q_{0.75}^{\text{wake}}$    | 1.0690, upper-quartile wake-score threshold                                  |
| ERA5    | $\varepsilon_{\text{sshf}}$ | 169,116, 10th percentile of  sshf  on the training split                     |
| ERA5    | $q_{0.95}^{\Delta}$         | 458,287.125, 95th percentile of $ \Delta \text{sshf} $ on the training split |

TABLE IV  
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**Table A5.** Active routing-loss weights in the reported corrected models.

| Weight                    | WTB   | ERA5 | Role                             |
|---------------------------|-------|------|----------------------------------|
| $\lambda_{\text{bal}}$    | 1000  | 0.01 | Expert-usage stabilization       |
| $\lambda_{\text{align}}$  | 5000  | 0.2  | Coarse regime alignment          |
| $\lambda_{\text{force}}$  | 10000 | 0    | Boundary-focused forcing         |
| $\lambda_{\text{aux}}$    | 250   | 0    | Wake-sensitive auxiliary routing |
| $\lambda_{\text{smooth}}$ | 0.05  | 0.05 | Local routing coherence          |

### Statistical claim boundaries

Table A6 records the reviewer-facing statistical boundary used for wording. The RMSE price versus Graph WaveNet is FDR-significant, gate-alignment deltas versus the full physics-aligned MoE are positive but not FDR-significant, and boundary-window quantile reserve comparisons have bootstrap intervals crossing zero. These tests are why the main text uses price, diagnostic, and bounded-claim language rather than forecast- or reserve-superiority wording.

TABLE V  
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**Table A6.** Statistical claim boundaries used for reviewer-facing wording.

| Claim                                           | Estimate | 95% CI             | $p_{\text{BH}}$ | Wording consequence                                                  |
|-------------------------------------------------|----------|--------------------|-----------------|----------------------------------------------------------------------|
| Accuracy price vs Graph WaveNet                 | 9.029    | —                  | <0.001          | Boundary router is significantly worse on headline RMSE              |
| Boundary-band accuracy price                    | 17.758   | —                  | 0.001           | The price also appears in the boundary window                        |
| Gate NMI vs full physics-aligned MoE            | 0.040    | [0.008, 0.071]     | 0.066           | Positive but not FDR-significant; cite as bounded mechanism contrast |
| Gate ARI vs full physics-aligned MoE            | 0.035    | [-0.000, 0.078]    | 0.126           | Positive but not FDR-significant; avoid superiority wording          |
| Boundary quantile cost vs GWN physical bin      | -0.263M  | [-10.704M, 9.782M] | —               | CI crosses zero; reserve cost should remain a diagnostic claim       |
| Boundary quantile violation vs GWN physical bin | 0.003    | [-0.011, 0.020]    | —               | CI crosses zero; no universal reserve-policy optimality claim        |

### Outcome-channel sanity audit

Table A7 adds a bounded check for the shared-anchor concern. It does not use pitch-threshold labels to score the contrast: validation data define a wind-speed-bin power curve, and the test-set comparison is restricted to 9.5–11.5 m s<sup>-1</sup> boundary anchors with fine wind-bin adjustment. The result asks whether the recovered gate separates samples with different future active-power response, not whether it discovers a regime without anchors.

TABLE VI  
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**Table A7.** Outcome-channel sanity audit for the shared-anchor concern.

| Check                   | Five-seed summary       | Interpretation                                                      |
|-------------------------|-------------------------|---------------------------------------------------------------------|
| Boundary cells per seed | 3276 MPPT / 10465 pitch | 9.5–11.5 m s <sup>-1</sup> test anchors                             |
| Future mean power       | 100.0 ± 17.5 kW         | Pitch-gate minus MPPT-gate after wind-bin adjustment                |
| Power-curve residual    | 100.0 ± 17.5 kW         | Validation wind-bin power curve only; no pitch label used           |
| Future power ramp       | 81.8 ± 24.7 kW          | Mean late-horizon minus early-horizon power                         |
| Anchor-time Patv        | -44.1 ± 32.5 kW         | Current active power is not driving the same positive contrast      |
| Residual claim boundary | sanity check only       | Mitigates circularity concern; does not prove anchor-free discovery |

### External-site deployment-gate audit

Table A8 records the go/no-go interpretation used for the Kelmarsh/Penmanshiel evidence. Each row maps a failed or incomplete external signal to the deployment action that follows. The table therefore supports a bounded claim: the external experiments are a screening protocol for new wind farms, not evidence that the WTB gate can be transferred directly.

TABLE VII  
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**Table A8.** External-site deployment-gate audit and wording boundary.

| Gate                          | Observed external evidence                                                                     | Go/no-go rule                                                                                                        | Claim consequence                                                                        |
|-------------------------------|------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|
| Cross-site routing criterion  | 80/80 runs complete; mean NMI 0.4877 below threshold 0.50; mean ARI 0.5112                     | Held-out NMI $\geq$ 0.50 and balanced accuracy $\geq$ 0.50 after parameters are frozen                               | No-go for cross-farm router interpretation; cite as negative boundary-condition evidence |
| Local boundary recalibration  | Default test NMI 0.1324 $\rightarrow$ recalibrated 0.1491 (delta +0.0167)                      | Rated wind, pitch threshold, boundary band, and gate-map selected on calibration only                                | Local threshold transfer is insufficient; re-estimate before use                         |
| Small-window adaptation       | 40/40 routing runs adapted; chronological balanced accuracy 0.4787 below 0.50                  | Small calibration windows must still pass the frozen held-out routing criterion                                      | Calibration alone does not authorize external reserve use                                |
| Sensor and boundary support   | Penmanshiel-to-Kelmarsh leave-one pitch-feature coverage 0.0000 and effective boundary cells 0 | Pre-declared calibration window with enough boundary cells, active power, availability mask, and pitch/proxy overlap | No physical-router interpretation without observability                                  |
| External reserve-use decision | Upstream gates do not pass before reserve allocation is evaluated                              | Transition-window shortage and violation improve at acceptable reserve-energy cost                                   | Withhold gate-bin reserve use outside WTB; report a deployment protocol only             |

### Early-warning detection consequence

Table A9 reports the cell-count version of the label-degradation audit. Counts are turbine-time cells per seed inside the six-step MPPT-to-pitch window; they are not MWh, currency, or dispatch-cost estimates. The purpose is narrower: it shows how many early pitch-window cells the gate preserves when a threshold-label rule is delayed, incomplete, or noisy.

TABLE VIII  
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**Table A9.** Early-warning detection consequence under degraded threshold labels.

| Condition               | Gate detected / missed             | Degraded rule detected / missed     | Recovered cells  |
|-------------------------|------------------------------------|-------------------------------------|------------------|
| Delay, 1 step           | 710.6 $\pm$ 20.6 / 29.4 $\pm$ 20.6 | 485.0 $\pm$ 0.0 / 255.0 $\pm$ 0.0   | 225.6 $\pm$ 20.6 |
| Delay, 3 steps          | 710.6 $\pm$ 20.6 / 29.4 $\pm$ 20.6 | 282.0 $\pm$ 0.0 / 458.0 $\pm$ 0.0   | 428.6 $\pm$ 20.6 |
| Delay, 6 steps          | 710.6 $\pm$ 20.6 / 29.4 $\pm$ 20.6 | 145.0 $\pm$ 0.0 / 595.0 $\pm$ 0.0   | 565.6 $\pm$ 20.6 |
| 50% label availability  | 710.6 $\pm$ 20.6 / 29.4 $\pm$ 20.6 | 376.0 $\pm$ 10.4 / 364.0 $\pm$ 10.4 | 334.6 $\pm$ 28.1 |
| 25% label availability  | 710.6 $\pm$ 20.6 / 29.4 $\pm$ 20.6 | 179.0 $\pm$ 7.9 / 561.0 $\pm$ 7.9   | 531.6 $\pm$ 22.9 |
| Sensor noise, strongest | 710.6 $\pm$ 20.6 / 29.4 $\pm$ 20.6 | 482.8 $\pm$ 11.0 / 257.2 $\pm$ 11.0 | 227.8 $\pm$ 20.9 |

### Reserve-policy claim-boundary audit

Table A10 consolidates the reserve evidence used for wording. It separates the same-model boundary-window diagnostic from claims that the experiments do not support. Costs are

normalized reserve-energy proxy units, not currency, market prices, or security-constrained dispatch costs.

TABLE IX  
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**Table A10.** Reserve-policy claim-boundary audit.

| Boundary                           | Evidence                                                                                                                                                                          | Wording rule                                                                                                        |
|------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------|
| Same-model boundary reserve effect | At $\rho=10$ , gate-bin vs same-router global: Delta cost -3.55M, Delta viol. -0.0138, reserve +1.85M, shortage -0.54M.                                                           | Claim a same-predictor transition-window diagnostic; do not present this as a cross-backbone reserve win.           |
| Physical-bin quantile comparator   | Boundary/physical-bin 83.78M, viol. 0.0880; GWN/physical-bin 84.31M, viol. 0.0931; Boundary/gate-bin 84.58M, viol. 0.0900.                                                        | Say physical-bin baselines are competitive and sometimes lower-cost; avoid gate-bin optimality wording.             |
| Cost-ratio applicability           | $\rho=2$ inactive; $\rho=5$ to 10 lowers boundary cost and violation; $\rho=20$ narrows; $\rho=50$ favors global (+5.93M, viol. +0.0035).                                         | Claim moderate-cost transition-window value only; do not assert a universal shortage-penalty policy.                |
| Full-sample system value           | Full sample at $\rho=10$ : GWN/global 464.07M, viol. 0.0901; Boundary/gate-bin 481.36M, viol. 0.1214.                                                                             | Do not claim system-wide dispatch value or reserve superiority; keep the consequence bounded to the boundary slice. |
| Seed-level uncertainty             | Full-sample paired total-cost delta +17.28M, 95% CI [-52.09M, +87.95M], perm. $p=0.752$ .                                                                                         | Use bounded diagnostic language; do not cite the reserve audit as a statistically settled improvement.              |
| Operational scope                  | Costs are normalized reserve-energy proxy units from validation-frozen shortfall quantiles; OPF, unit commitment, delivery constraints, market clearing, and prices are excluded. | Use as a screening audit for reserve exposure, not as a market or security-constrained dispatch study.              |

### Engineering-unit reserve-value translation

Table A11 provides the engineering-unit translation of the main reserve audit. The conversion uses the WTB active-power unit (kW) and the cache time step ( $\Delta t = 1/6$  h), so reserve and shortage totals become rolling forecast-cell MWh-equivalent values. The EUR column is a scenario translation under an assumed reserve carrying cost of 100 EUR/MWh. It is included to make the operational scale legible, not to claim market settlement, OPF, unit commitment, or security-constrained dispatch value.

TABLE X  
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**Table A11.** Engineering-unit reserve-value translation at  $\rho = 10$ .

| Comparison                                        | $\Delta$ reserve<br>MWh-eq. | Avoided<br>shortage<br>MWh-eq. | $\Delta$ cost<br>MWh-eq. | $\Delta$ EUR at<br>100/MWh | Wording                                                                                                            |
|---------------------------------------------------|-----------------------------|--------------------------------|--------------------------|----------------------------|--------------------------------------------------------------------------------------------------------------------|
| Boundary gate-bin<br>vs same-router<br>global     | +1846.9                     | +539.8                         | -3551.4                  | -355k                      | Use as<br>bounded<br>boundary-<br>window<br>value, not<br>cross-<br>backbone<br>superiori-<br>ty.                  |
| Boundary gate-bin<br>vs GWN<br>physical-bin       | +2319.2                     | +205.6                         | +262.9                   | +26k                       | Shows<br>gate-bin is<br>close to a<br>strong<br>physical-<br>bin<br>compara-<br>tor; not a<br>lower-cost<br>claim. |
| Boundary gate-bin<br>vs GWN global full<br>sample | +3148.2                     | -1413.5                        | +17283.2                 | +1728k                     | Blocks<br>system-<br>wide<br>dispatch or<br>full-<br>sample<br>reserve-<br>superiority<br>wording.                 |

MWh-eq. denotes forecast-cell MWh-equivalent accounting from kW active-power shortfall and  $\Delta t = 1/6$  h. EUR values are scenario translations under an assumed reserve carrying cost of 100 EUR/MWh; they are not market-settlement, OPF, or unit-commitment results.